

Gennoor Tech

TRAIN. INNOVATE. BUILD.

GENNOOR ACADEMY · SELF-PACED ·
36 MIN

AI for Energy & Utilities

A ~36-minute regulator-aware playbook for utility execs, grid operators, generation planners, digital transformation heads. 5 plays, operator-in-the-loop, OT+IT+AI cyber, 18-month roadmap.

8

CHAPTERS

36 min

DURATION

Advanced

LEVEL

FREE

CERTIFICATE

Course Overview

For utility executives, grid operators, generation planners, and heads of digital transformation in energy companies. AI must augment operational reliability, not threaten it — grid stability and human safety are not negotiable. This is different from most enterprise AI.

FORMAT: Self-paced · 8 chapters · audio-narrated slides · interactive end-of-course builder · free certificate on completion

Who Is This For?

1

Business + technology
leaders

2

Function heads +
transformation owners

3

L&D + internal trainers

4

AI-curious managers

Learning Outcomes

- 1 AI must augment operational reliability, not threaten it — grid stability and worker safety are non-negotiable
- 2 5 high-value plays — demand forecasting · grid management · generation optimisation · predictive maintenance · customer ops
- 3 Keep operator-in-the-loop on grid management — NERC + FERC + PUC oversight makes it non-negotiable
- 4 Run renewables forecasting and battery dispatch optimisation — where AI directly impacts revenue at scale
- 5 Use AI to extend asset inspection intervals — never to eliminate them (OSHA + worker safety)
- 6 Hold the customer disconnection line — AI flags, humans decide
- 7 Treat OT + IT + AI cybersecurity as one problem — 4 attack vectors, 3 defensive patterns
- 8 18-month rollout with 4 trust trip-wires and an interactive builder for your CIO, COO, or board

8-Chapter Curriculum

Self-paced, audio-narrated slide format. Each chapter ends with a "Monday move" the learner can apply that week.

01

The energy + utilities landscape

CHAPTER 1 · 5 MIN · NARRATED + INTERACTIVE

- AI must augment reliability, not threaten it
- 5 high-value plays
- regulator frame (NERC + FERC + PUC + EU AI Act + Network Code Cyber).

02

Demand forecasting

CHAPTER 2 · 5 MIN · NARRATED + INTERACTIVE

- Most mature play
- 5-15% accuracy lift
- 3 architectures (GBM
 - neural
 - hybrid)
- 3 failure modes

03

Grid management

CHAPTER 3 · 5 MIN · NARRATED + INTERACTIVE

- 3 use cases (asset health
 - congestion
 - outage response)
- operator-in-the-loop is non-negotiable
- 3 failure modes.

04

Generation optimisation

CHAPTER 4 · 5 MIN · NARRATED + INTERACTIVE

- Why renewables need AI more than thermal
- 3 use cases (wind
 - solar
 - battery dispatch)
- the curtailment fairness question.

05

Predictive maintenance for T+D assets

CHAPTER 5 · 5 MIN · NARRATED + INTERACTIVE

- Tens of millions deferred capital
- 3 use cases (transformer
 - switchgear
 - line)
- safety constraint: AI extends inspections, never eliminates.

06

Customer operations

CHAPTER 6 · 5 MIN · NARRATED + INTERACTIVE

- 3 use cases (outage bills
- advice)
- the disconnect line (AI never autonomous)
- vulnerable customer protection.

07

Cybersecurity for AI in critical infrastructure

CHAPTER 7 · 5 MIN · NARRATED + INTERACTIVE

- OT+IT+AI is one problem
- 4 attack vectors (poisoning
- evasion
- exfiltration
- supply chain)
- 3 defensive patterns.

08

Making it stick: your energy AI roadmap

CHAPTER 8 · 6 MIN · NARRATED + INTERACTIVE

- 18-month rollout (foundations
- scale
- operating discipline)
- 4 trust trip-wires
- interactive Markdown builder.